

HUJI Research Monitor

Weekly Digest · Week 28 · July 06, 2026

EXECUTIVE SUMMARY

This week's digest highlights groundbreaking advancements across several high-impact fields. We feature pioneering cell and gene therapies targeting severe autoimmune diseases and neurodegenerative disorders, alongside novel drug candidates for aggressive cancers and regenerative medicine. Significant opportunities also emerge in precision diagnostics, medical devices for infection control and ophthalmology, and AI-driven platforms accelerating drug discovery, collectively addressing critical unmet needs in global health and industry.

#1

B cell Maturation Antigen Targeted Chimeric Antigen Receptor T Cell Therapy for Refractory Systemic Lupus Erythematosus and Systemic Sclerosis.

37/50

Main Researcher: Chamutal Gur

[Immunology] [Clinical] [Synthetic Biology]

"Pioneering CAR T-cell therapy offers hope for severe autoimmune diseases like lupus, targeting high-value chronic conditions."

WHY NOW

Represents a significant paradigm shift from oncology into large, underserved autoimmune markets, attracting major pharma interest in chronic disease management. Early clinical data shows promising efficacy despite known risks associated with CAR T-cell therapies.

COMMERCIAL ANGLE

Developing specialized CAR T-cell therapies for autoimmune indications represents a massive shift from oncology toward high-value chronic disease management markets.

<https://europepmc.org/article/MED/42098533>

#2

Pro-repair properties of a human embryonic stem cell-derived astrocyte cell therapy in demyelinating disorders.

36/50

Main Researcher: Tamir Ben-Hur

[Neuroscience] [Immunology] [Clinical] [Synthetic Biology]

"An off-the-shelf stem cell therapy to repair myelin in demyelinating disorders, including Multiple Sclerosis."

WHY NOW

Addresses a critical unmet need for remyelination therapies in MS and other neurological conditions, offering a scalable, clinical-grade solution for a multi-billion dollar market. This therapy's pro-repair properties could revolutionize treatment paradigms.

COMMERCIAL ANGLE

Development of a scalable, off-the-shelf cell therapy for remyelination in Multiple Sclerosis and other demyelinating neurological disorders.

<https://pubmed.ncbi.nlm.nih.gov/42276062/>

#3

33/50

The effect of a proprietary chlorine-stabilizing surface-coating formulation on healthcare environment disinfection in multidrug-resistant bacteria isolation rooms.

Main Researcher: Oren Zimhony

[Medical Device] [Materials] [Clinical]

"Proprietary surface coating boosts hospital disinfection, battling multidrug-resistant bacteria in high-touch environments."

WHY NOW

Offers a highly scalable and durable solution to enhance infection control, reducing healthcare-associated infections and combating the global threat of antimicrobial resistance. This technology has immediate application in hospitals and long-term care facilities.

COMMERCIAL ANGLE

This technology offers a scalable solution for hospitals and long-term care facilities to enhance infection control protocols and reduce the transmission of multidrug-resistant organisms through high-durability antimicrobial coatings.

<https://pubmed.ncbi.nlm.nih.gov/42374536/>

#4

33/50

Recombinant Human Amelogenin Protein Enhances Healing of Osteochondral Injury in a Goat Model.

Main Researcher: Amir Haze

[Medical Device] [Clinical] [Synthetic Biology] [Proteomics]

"Recombinant human protein promotes robust cartilage and bone healing for osteoarthritis and traumatic joint injuries."

WHY NOW

Taps into the massive and growing market for regenerative medicine solutions in orthopedics, offering a high-value biologic with significant clinical potential for joint repair. This therapy addresses a major cause of chronic pain and disability.

COMMERCIAL ANGLE

Development of a high-value regenerative medicine biologic for treating osteoarthritis and traumatic joint injuries in humans.

<https://pubmed.ncbi.nlm.nih.gov/41925697/>

#5

APOBEC mutational signatures predict immune checkpoint inhibitor benefit in TMB-low metastatic NSCLC independent of PD-L1 status.

32/50

Main Researcher: N Peled

[Diagnostics] [Genomics] [Immunology] [Software/AI]

"AI-driven diagnostic identifies new responders to immunotherapy in lung cancer, expanding market reach."

WHY NOW

Fills a critical gap in patient stratification for immune checkpoint inhibitors, enabling broader application and increasing market opportunities for existing blockbuster drugs. This companion diagnostic provides a new tool for precision oncology.

COMMERCIAL ANGLE

Development of a companion diagnostic test using machine learning to identify APOBEC signatures in TMB-low lung cancer patients to expand the market for immune checkpoint inhibitors.

<https://pubmed.ncbi.nlm.nih.gov/42378718/>

#6

Targeting primary and metastatic ovarian cancer with a peptide derived from the human NAF-1/CISD2 protein.

29/50

Main Researcher: Rachel Nechushtai

[Drug Discovery] [Clinical] [Immunology]

"Novel peptide targets and inhibits primary and metastatic ovarian cancer, offering a first-in-class therapeutic."

WHY NOW

Ovarian cancer remains an aggressive malignancy with high unmet needs; this discovery provides a promising and differentiated therapeutic candidate for pharmaceutical licensing. The peptide's specificity could lead to fewer off-target effects.

COMMERCIAL ANGLE

The peptide represents a high-potential therapeutic candidate for a first-in-class oncology drug, suitable for licensing to pharmaceutical companies focusing on precision cancer therapies or protein-protein interaction inhibitors.

<https://pubmed.ncbi.nlm.nih.gov/42259134/>

#7

Breath-Based Detection of Oral Diseases Using Sensors and Machine Learning.

28/50

Main Researcher: Yael Houri-Haddad

[Medical Device] [Diagnostics] [Software/AI]

"Rapid, non-invasive breathalyzer for dental clinics provides objective diagnostics for various oral diseases."

WHY NOW

Meets the demand for point-of-care diagnostic tools in dentistry, offering a novel, non-invasive method to objectively screen for periodontal health and other oral conditions. This technology could significantly improve early detection and management.

COMMERCIAL ANGLE

Development of a non-invasive, rapid-screening handheld breathalyzer for dental clinics to provide objective diagnostics for periodontal health.

<https://pubmed.ncbi.nlm.nih.gov/42379259/>

#8

RadD from *Fusobacterium nucleatum* engages NKp46 to promote antitumor cytotoxicity.

28/50

Main Researcher: Ofer Mandelboim

[Immunology] [Drug Discovery] [Clinical]

"Bacterial protein activates NK cells to combat tumors, paving the way for novel cancer immunotherapies."

WHY NOW

Capitalizes on the booming field of immuno-oncology by offering a new mechanism to activate natural killer cells, a key component of the innate immune system, against cancer. This approach could lead to new biomimetic immunotherapies.

COMMERCIAL ANGLE

Development of RadD-based NK cell agonists or biomimetic immunotherapies to enhance natural killer cell cytotoxicity in oncology.

<https://europepmc.org/article/MED/42065370>

#9

Aging-associated modulation of UFMylation impairs proteostasis in *C. elegans*.

27/50

Main Researcher: Ehud Cohen

[Drug Discovery] [Neuroscience] [Proteomics]

"Targeting protein modification extends lifespan and prevents protein buildup, potentially treating neurodegenerative diseases."

WHY NOW

Offers a novel, disease-modifying pharmacological strategy for major neurodegenerative disorders like Alzheimer's and Parkinson's, representing a multi-billion dollar market with high unmet needs. This fundamental discovery opens new therapeutic avenues.

COMMERCIAL ANGLE

Targeting the UFMylation pathway offers a novel pharmacological strategy for developing disease-modifying therapies for neurodegenerative disorders like Alzheimer's and Parkinson's.

<https://pubmed.ncbi.nlm.nih.gov/42056101/>

#10

Targeting Temozolomide-Resistant Glioblastoma: Therapeutic Potential of Neuronal Nitric Oxide Synthase Inhibitor.

26/50

Main Researcher: Haitham Amal

[Drug Discovery] [Neuroscience] [Clinical]

"Nitric oxide inhibitor enhances glioblastoma chemotherapy, overcoming resistance in deadly brain tumors."

WHY NOW

Addresses the urgent need for improved treatments for glioblastoma, offering a high-value adjuvant therapy that extends the efficacy of standard-of-care chemotherapy against drug resistance. This could significantly improve patient outcomes for a devastating cancer.

COMMERCIAL ANGLE

Development of a high-value adjuvant therapy to extend the clinical utility of existing standard-of-care chemotherapy for resistant brain tumors.

<https://pubmed.ncbi.nlm.nih.gov/42374654/>

#11

Strategic Template Filtering Accelerates Fragment-Based Peptide Docking.

26/50

Main Researcher: Ora Schueler-Furman

[Drug Discovery] [Software/AI] [Proteomics]

"High-efficiency AI algorithm accelerates peptide docking, optimizing drug discovery for novel therapeutics."

WHY NOW

Provides a crucial computational advantage for pharmaceutical companies developing peptide-based drugs, streamlining design and significantly reducing R&D costs and time-to-market. This software is a valuable tool for AI-driven drug discovery platforms.

COMMERCIAL ANGLE

Licensing or integrating this high-efficiency docking algorithm into AI-driven drug discovery platforms to accelerate the design of peptide-based therapeutics.

<https://pubmed.ncbi.nlm.nih.gov/42299509/>

#12

Safety and Efficacy of Direct Selective Laser Trabeculoplasty in Patients With Ocular Hypertension or Open Angle Glaucoma.

26/50

Main Researcher: David Zadok

[Medical Device] [Clinical]

"Laser therapy for glaucoma reduces eye pressure, offering a safe and effective first-line alternative to drops."

WHY NOW

Expands the market for ophthalmic surgical devices by offering a non-pharmacological, first-line treatment for glaucoma and ocular hypertension, reducing patient burden and improving adherence. This could become a preferred treatment option.

COMMERCIAL ANGLE

The findings support the commercialization and expanded clinical adoption of laser-based ophthalmic surgical devices as a first-line alternative to traditional glaucoma eye drops.

<https://europepmc.org/article/MED/42148860>